SOFTWARE ENGINEERING

LAB ASSIGNMENT 09

UML Component Diagram

Digital Wallet Management System (DWMS)

SUBMITTED BY — TEAM 5

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1. Introduction to Component Diagrams

A UML Component Diagram represents the modular organization of a software system and the dependencies among its components. It provides a structural view of services, user interfaces, data stores, and external systems. Component diagrams help teams reason about separation of concerns, integration boundaries, maintainability, and the impact of changing one module.

1.1 Objectives

- Decompose the DWMS into components aligned with the layered architecture in the SRS.
- Show the principal dependencies among UI, authentication, wallet/transaction logic, persistence, notifications, and external providers.
- Distinguish internal components from external bank/payment and OTP/notification services.

2. Architectural Basis from the SRS

The SRS defines four logical layers: presentation (web/mobile client), application (authentication, wallet, transaction, payment, notification services), data (users, wallets, transaction ledger, audit logs), and integration (bank APIs, payment gateway, OTP/notification services). The diagram preserves these responsibilities while showing representative components inside the DWMS boundary.

2.1 Component Responsibilities

- Web/Mobile UI: registration/login, wallet dashboard, funding, transfer, payment, history, and notifications screens.
- API/Auth Component: exposes client-facing operations and enforces authentication/session and authorization checks.
- Wallet & Transaction Component: applies wallet, transfer, and payment business rules.
- Persistence Component and DWMS Database: stores users, wallets, ledger entries, and audit logs durably.
- Audit & Admin Component: supports user/transaction monitoring and records administrative actions.
- Notification Component: creates account and transaction notices and calls the configured delivery provider.
- External providers: bank/payment gateway processes supported financial operations; OTP/notification provider delivers verification codes and notices.

UML DIAGRAM

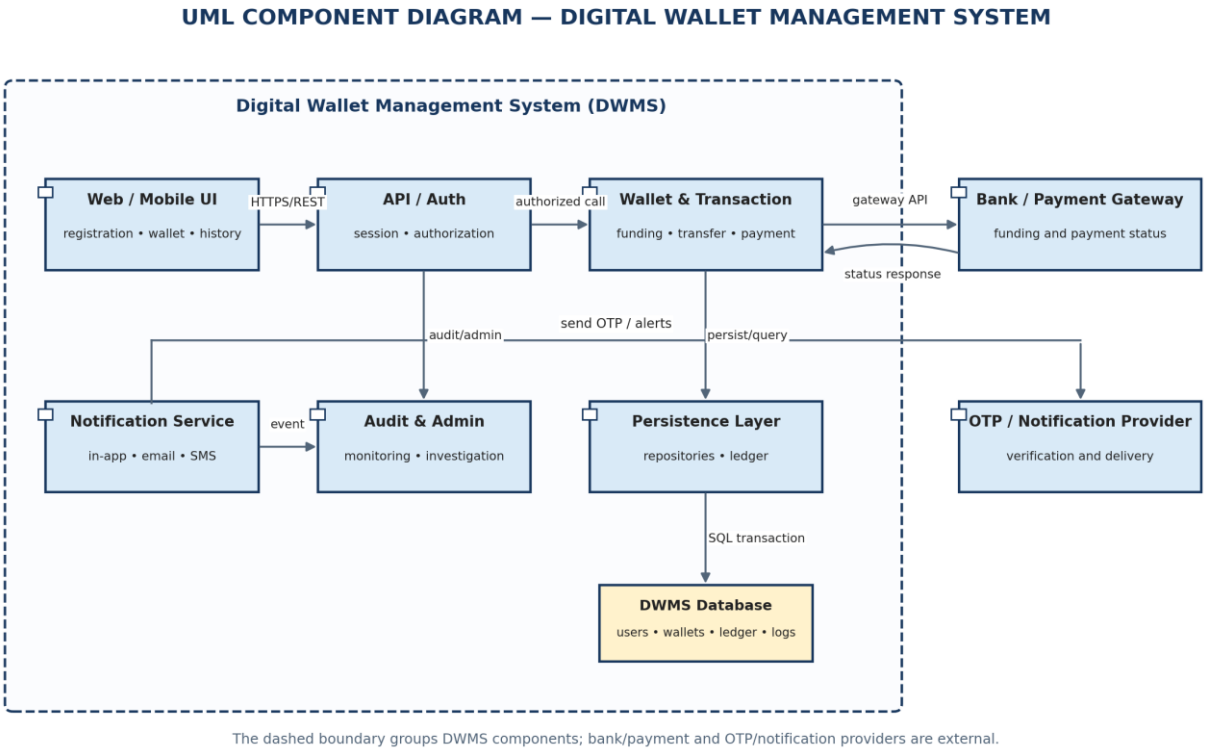


Figure 1. UML component diagram for the Digital Wallet Management System.

3. Dependency and Interface Explanation

- Client requests reach the API/Auth component over HTTPS/REST.
- Authorized requests are passed to wallet and transaction business logic.
- Business components use the persistence layer to read/write wallet and ledger data.
- Payment operations communicate with the external bank/payment gateway and use its response to determine final status.
- Notification and verification events are sent to the configured OTP/notification provider.
- Administrative monitoring is separated from ordinary customer operations and its actions are audited.

4. Design Justification

The component boundaries follow the SRS's modular, layered architecture constraint. Separating presentation, business rules, persistence, and integration improves maintainability and testability. It also makes security controls easier to apply at service boundaries and avoids embedding external bank/payment implementation details inside the DWMS.

Conclusion

The component diagram translates the SRS's layered architecture into identifiable software modules and communication paths. It illustrates how core wallet functions depend on persistence and external services while

keeping administrative monitoring and notifications as distinct responsibilities. This structure supports the stated goals of security, reliability, scalability, and maintainability.